

Experiments List of Digital Image Processing

1. Basic Image Manipulation & Enhancement

- **Image Acquisition and Display:** Reading, displaying, and converting image types (RGB to Gray, Binary).
- **Pixels Relationships:** Implementation of Relationships between Pixels
- **Intensity Transformations:** Image negative, log transformation, and power-law (Gamma) transformation.
- **Contrast Enhancement:** Contrast stretching and piecewise linear transformation.
- **Histogram Processing:** Computing histograms and implementing histogram equalization.
- **Spatial Filtering (Smoothing/Sharpening):**
 - **Smoothing:** Averaging (mean) filters and median filters to remove noise.
 - **Sharpening:** Laplacian filter and high-boost filtering.
- **Bit Plane Slicing:** Displaying individual bit planes of an image.

2. Frequency Domain Processing

- **Fast Fourier Transform (FFT):** Computing 1D and 2D DFT of an image.
- **Frequency Domain Filtering:**
 - **Low Pass Filters:** Ideal, Butterworth, and Gaussian low pass filters.
 - **High Pass Filters:** Ideal, Butterworth, and Gaussian high pass filters.

3. Image Segmentation and Morphological Processing

- **Edge Detection:** Implementing Sobel, Prewitt, and Roberts cross-gradient operators.
- **Segmentation:** Point, line, and edge detection, as well as thresholding.
- **Morphological Operations:** Dilation, erosion, opening, closing, and top-hat transformations.
- **Color Image Processing:** Analysis of images with different color models (RGB, HSV, CMY).

4. Image Restoration and Compression

- **Noise Modeling:** Implementing Gaussian, Rayleigh, and Salt & Paper noise.
- **Image Restoration:** Restoration using various filters.
- **Compression:** Image compression techniques like Huffman coding or Discrete Cosine Transform (DCT).

Write a Python Program for reading an image and Convert it into grayscale using formula

$$* \text{img_gray1} = 0.299R + 0.587G + 0.114B$$

$$* \text{img_gray2} = (R + G + B) / 3$$

Compare with Built-in function

Implementation of Relationships of Pixel in an image 5x5.

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25